

Table R1: Efficiency comparison of Hi-Q against baselines.

Method	Latency (s)	Online Graph		LLM Calls			Tokens	
		Traversal Latency	Subquestion	HotpotQA	MuSiQue	2WikiMHQA	Input	Output
IRCOT	2.6	–	1.2	1.1	1.3	1.4	18,700	58
PropRAG	19.7	17.5	–	1	1	1	1,411	88
Hi-Q	13.5	–	1.1	2.4	8.5	6.1	12,343	558

Table R2: Comparison of Hi-Q latency against Single Retrieval with GPT-5.

Method	Latency	F1	Recall@5	2hop F1	3hop F1	4hop F1
Single Retrieval w/ GPT-5	18.4	50.0	50.0	60.0	67.0	0.0
Hi-Q	13.5	66.7	65.0	53.0	100.0	50.0

Table R3: Full-corpus MuSiQue evaluation.

Method	EM	F1	Recall@2	Recall@5
IRCOT	25.8	38.6	44.4	57.9
PropRAG	30.2	43.7	46.3	60.7
Hi-Q	36.3	48.7	48.4	64.6

Table R4: Ablation of the verification module stratified by reasoning complexity.

	Overall	2hop	3hop	4hop
Hi-Q	57.2	64.3	55.2	39.2
w/o round-trip	56.5	63.7	54.4	38.2

Table R5: Comparison against dedicated decomposition baselines.

Method	Recall
Least-to-Most (ICLR 2023)	13.4
Self-Ask (EMNLP 2023)	15.6
TRQA (AAAI 2024)	26.8
Hi-Q	29.3

Table R6: Controlled experiment: IRCOT with gold decomposition.

Method	EM	F1	Recall@2	Recall@5
IRCOT	37.5	51.2	54.4	71.8
IRCOT w/ gold decomposition	42.8	58.4	63.8	84.8

Table R7: Decomposition trigger rate and round-trip repair frequency across datasets.

Dataset	Decomposition Trigger Rate (%)	Round-trip Repair Frequency (%)
HotpotQA	10.7	18.8
MuSiQue	45.2	14.8
2WikiMHQA	48.6	17.0

Table R8: Manual analysis of 100 randomly sampled $a = \perp$ triggers.

Trigger Cause	Proportion (%)	Description
Granularity Mismatch	66	–
Annotation Error	14	Dataset-level flaws (e.g., missing gold info)
Malformed Subquery	10	Failed to incorporate prior context
False Abstention	7	Model had sufficient info but returned null
Reader Failure	3	Failed to recognize information

Table R9: Performance with Llama-3.3-70B as the backbone LLM (F1).

	HotpotQA	MuSiQue	2Wiki	Avg
IRCOT	72.0	40.7	78.3	63.7
PropRAG	76.0	53.6	74.3	68.0
Hi-Q	78.0	56.6	78.9	71.2